

Testing of Hypothesis Foundations

Null alternative hypotheses, Type I II errors, p-values, Z-tests for means and proportions.

1 Principles of Hypothesis Testing

Hypothesis testing is a structured statistical framework used to validate claims about population parameters using sample data.

The Courtroom Analogy

In a criminal trial, the defendant is assumed innocent until proven guilty beyond a reasonable doubt.

- **Null Hypothesis (H_0):** Defendant is innocent (status quo / no effect).
- **Alternative Hypothesis (H_1):** Defendant is guilty (the claim we seek to prove).
- **Burden of Proof:** We only reject H_0 if sample evidence is overwhelmingly unlikely under H_0 .

2 Type I and Type II Errors

Because sample data carries randomness, statistical decisions can lead to two types of errors:

- **Type I Error (α):** Rejecting H_0 when H_0 is actually true (False Alarm).
- **Type II Error (β):** Failing to reject H_0 when H_1 is true (Missed Detection).
- **Power of the Test ($1 - \beta$):** Probability of correctly rejecting a false H_0 .

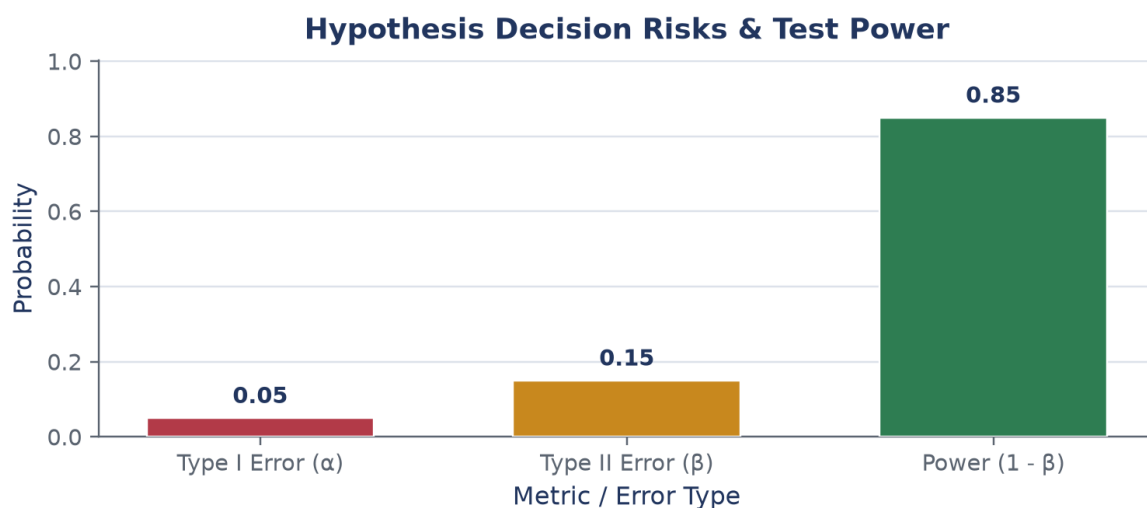


Figure: Trade-offs between Type I Error α , Type II Error β , and Test Power $1 - \beta$.

3 One-Tailed vs Two-Tailed Z-Tests

Depending on the alternative hypothesis, rejection regions sit in one tail or split across both tails.

Two-Tailed Test: $H_0 : \mu = \mu_0$ vs $H_1 : \mu \neq \mu_0$
 Right-Tailed Test: $H_0 : \mu \leq \mu_0$ vs $H_1 : \mu > \mu_0$
 Left-Tailed Test: $H_0 : \mu \geq \mu_0$ vs $H_1 : \mu < \mu_0$

Two-Tailed Hypothesis Test ($\alpha = 0.05$, Critical Z = ± 1.96)

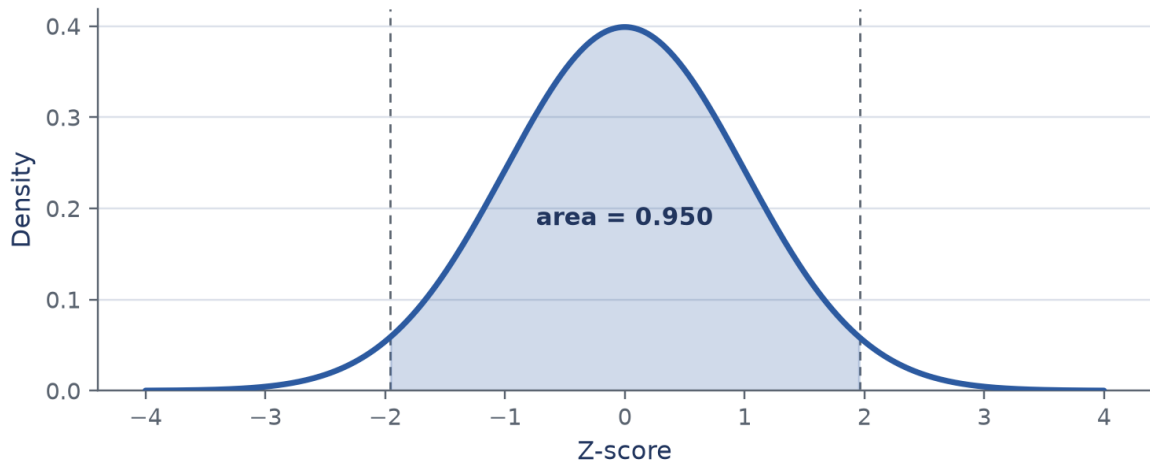


Figure: Two-tailed Z-test with critical bounds $Z = \pm 1.96$ at significance level $\alpha = 0.05$.

Right-Tailed Hypothesis Test ($\alpha = 0.05$, Critical Z = 1.645)

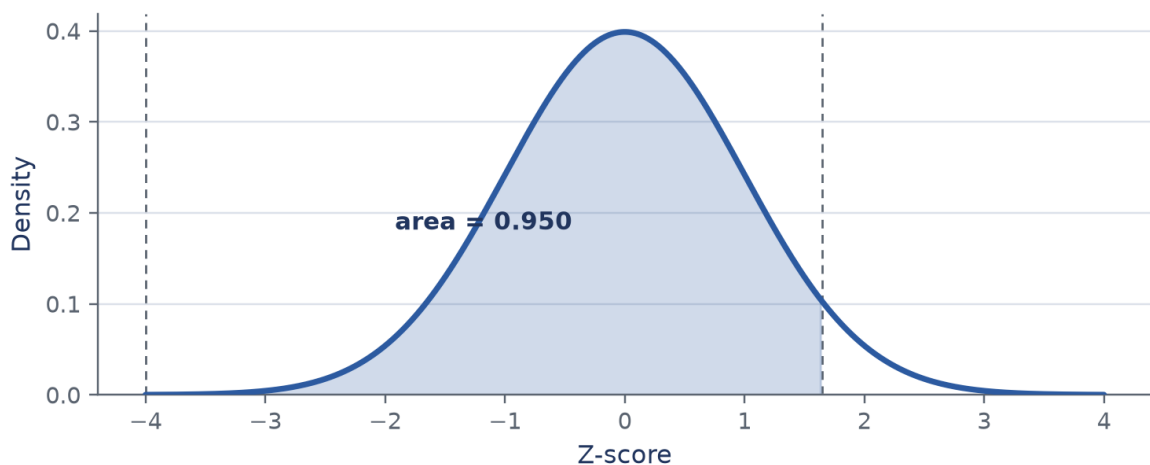


Figure: Right-tailed Z-test with critical bound $Z = +1.645$ at significance level $\alpha = 0.05$.

Slide Example: Testing Difference of Two Population Means

Problem Statement: Two machine processes produce steel rods. A sample of $n_1 = 50$ rods from Machine 1 yields mean breaking strength $\bar{X}_1 = 38$ kg with $\sigma_1 = 4$ kg. A sample of $n_2 = 60$ rods from Machine 2 yields $\bar{X}_2 = 35$ kg with $\sigma_2 = 5$ kg. Test if there is a significant difference between the true population means at $\alpha = 0.05$.

Step 1: State Hypotheses:

$$H_0 : \mu_1 - \mu_2 = 0 \quad \text{vs} \quad H_1 : \mu_1 - \mu_2 \neq 0$$

Step 2: Determine Critical Value: For a two-tailed Z-test at $\alpha = 0.05$, critical values are $Z_{\text{crit}} = \pm 1.96$.

Step 3: Calculate Standard Error of Difference:

$$SE_{\bar{X}_1 - \bar{X}_2} = \sqrt{\frac{\sigma_1^2}{n_1} + \frac{\sigma_2^2}{n_2}} = \sqrt{\frac{4^2}{50} + \frac{5^2}{60}} = \sqrt{\frac{16}{50} + \frac{25}{60}} = \sqrt{0.320 + 0.417} = \sqrt{0.737} \approx 0.858$$

Step 4: Compute Z Test Statistic:

$$Z_{\text{cal}} = \frac{(\bar{X}_1 - \bar{X}_2) - 0}{\text{SE}} = \frac{38 - 35}{0.858} = \frac{3}{0.858} \approx 3.50$$

Step 5: Make Decision: Since $Z_{\text{cal}} = 3.50 > 1.96$, we reject H_0 . There is strong evidence of a significant difference between the two machines.

Slide Example: Test of Significance for Single Proportion

Problem Statement: A phone survey claims that 60% ($p_0 = 0.60$) of voters favor candidate A. In a sample of $n = 400$ voters, $X = 220$ favor candidate A ($\bar{p} = 220/400 = 0.55$). Test if candidate support is significantly lower than claimed at $\alpha = 0.05$.

Step 1: State Hypotheses:

$$H_0 : p \geq 0.60 \quad \text{vs} \quad H_1 : p < 0.60 \quad (\text{Left-Tailed Test})$$

Step 2: Determine Critical Value: For left-tailed test at $\alpha = 0.05$, $Z_{\text{crit}} = -1.645$.

Step 3: Calculate Standard Error:

$$\text{SE}_{\bar{p}} = \sqrt{\frac{p_0(1-p_0)}{n}} = \sqrt{\frac{0.60 \times 0.40}{400}} = \sqrt{\frac{0.24}{400}} = \sqrt{0.0006} \approx 0.0245$$

Step 4: Compute Z Test Statistic:

$$Z_{\text{cal}} = \frac{\bar{p} - p_0}{\text{SE}_{\bar{p}}} = \frac{0.55 - 0.60}{0.0245} = \frac{-0.05}{0.0245} \approx -2.04$$

Step 5: Make Decision: Since $Z_{\text{cal}} = -2.04 < -1.645$, we reject H_0 . Support is significantly lower than 60%.

p-Value Decision Rule

Always compare p -value directly to significance level α :

- If $p\text{-value} \leq \alpha$, **Reject** H_0 ("If p is low, H_0 must go").
- If $p\text{-value} > \alpha$, **Fail to reject** H_0 .

Summary Checklist

- **H_0 vs H_1 :** H_0 assumes equality/no effect; H_1 holds the claim to prove.
- **Type I Error α :** False alarm probability set by the analyst.
- **Decision Rule:** Reject H_0 when $|Z_{\text{cal}}| > Z_{\text{crit}}$ or $p\text{-value} \leq \alpha$.